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Effect of orientation on forced convection heat transfer from a heated cone in Bingham plastic fluids

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ABSTRACT

Forced convection heat transfer from an isothermal heated cone (apex up and down) in Bingham plastic fluids has been studied numerically. Extensive results are presented here in terms of isotherm and Nusselt number over ranges of conditions as: Reynolds number, $1 \le Re \le 100$, Prandtl number, $10 \le Pr \le 100$, Bingham number, $0 \le Bn \le 100$ and cone angle, $20^{\circ} \le \alpha \le 150^{\circ}$. The rate of heat transfer is influenced mainly by the values of *Re*, *Pr* and *Bn* and to a lesser extent by the cone angle and orientation. The present results are consolidated in terms of the modified Reynolds (Re^*) and Prandtl numbers (Pr^*) incorporating the role of Bingham number.

1. Introduction

Most substances of multiphase nature and/or structured fluids like emulsions, foams and suspensions exhibit viscoplastic fluid behavior. This class of fluids deviates from the usual Newtonian fluids in so far that such a substance exhibits a fluid-like behavior only when the applied stress exceeds the fluid yield stress. Thus the flow domain consists of both fluid-like (yielded) and solid-like (unyielded) regions. The conduction-dominated heat transfer occurring in the latter is often the overall limiting step in such applications thereby making their heating, cooling and homogenization rather difficult. Consequently, much research effort has been directed at studying the convective momentum and heat transfer in such fluids in internal and external flows, past objects of two and three dimensional shapes, e.g., see Chhabra [1], though much attention has been accorded to the case of a sphere [2,3] or a spheroid [4]. Also, most of these studies deal with the fluid mechanical aspects and the analogous heat transfer phenomenon has been studied scantly. The dependence of the average Nusselt on the Bingham number varies from one flow regime to another. Thus in the forced convection regime Bingham number augments the overall heat transfer due to the progressive thinning of the fluid-like layer next to the heated object. On the other hand, the fluid yield stress (Bingham number) has deleterious effect on heat transfer in the free convection regime. Naturally, the velocity and temperature fields in the vicinity of a cone are influenced by the cone angle and orientation and this work is aimed at studying their influence on heat transfer in the forced convection heat transfer regime. In addition, to the fundamental significance of this geometry, it has relevance in the measurement of flow and rheological properties of soils [5,6], soft solids encountered in food and personal care product testing where the cone penetration test constitutes a standard quality control method [7,8]. Conversely, this approach can also be extended to establish the effect of temperature on the yield strength and other rheological characteristics by using a heated cone and measuring the rate of heat transfer. As noted elsewhere [9], there is a very limited information available on the drag and the Nusselt number behavior of a cone, even in Newtonian fluids [10,11]. While the drag behavior of a cone translating in Bingham plastic fluids has been explored in few studies [9,12], there are no corresponding results on heat transfer available in the literature. The present study is thus aimed at analyzing the effects of Reynolds number, ($1 \le Re \le 100$), Prandtl number ($10 \le Pr \le 100$), Bingham number, ($0 \le Bn \le 100$) and cone angle, ($20^{o} \le \alpha \le 150^{o}$) and two orientations (apex up and down) on heat transfer from an isothermal cone.

2. Problem formulation

Fig. 1 shows the two orientations of the cone (diameter *D*; temperature $T_s > T_\infty$) exposed to a uniform stream of a Bingham plastic medium ($U_z = U_\infty$ and at temperature T_∞). The flow is expected to be steady and axisymmetric; also the physical properties of the fluid (k, C, μ_B, τ_0, and ρ_∞) are assumed to be constant as $\Delta T = T_s - T_\infty = 5K$ is small in this study. Similarly, the small value of $Br \sim 0.002$ justifies to disregard the viscous dissipation effects in the energy equation. Thus for incompressible, steady and laminar flow, the velocity and the temperature fields in the fluid are governed by the continuity, momentum and energy equations, respectively written in their non-

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Nomenclature

Bn	Bingham number ($=\tau_0 D/\mu_B U_\infty$), dimensionless
Br	Brinkman number ($=\mu_B U_\infty{}^2/k(T_s - T_\infty)$), dimensionless
C	heat capacity of fluid, $J{\cdot}kg^{-1}{\cdot}K^{-1}$
D	diameter of cone, m
D_∞	diameter of the computational domain, m
h	heat transfer coefficient, $W{\cdot}m^{-2}{\cdot}K^{-1}$
j_H	Colburn-j_H factor for heat transfer, dimensionless
k	thermal conductivity of fluid, $W{\cdot}m^{-1}{\cdot}K^{-1}$
m	regularization parameter, dimensionless
n_s	unit vector normal to the surface of the cone, dimensionless
Nu_{avg}	average Nusselt number, dimensionless
Nu_L	Local Nusselt number, dimensionless
Pe	Peclet number ($= Re \cdot Pr$), dimensionless
Pr	Prandtl number ($=C\mu_B/k$)dimensionless
Pr^*	modified Prandtl number ($= Pr(1 + Bn)$), dimensionless
Re	Reynolds number($=\rho D U_\infty/\mu_B$), dimensionless
Re^*	modified Reynolds number ($= Re/(1 + Bn)$), dimensionless
S	surface area of cone, m^2
T	temperature of fluid, K
T_s	cone surface temperature, K
T_∞	inlet fluid temperature, K
U_∞	free stream velocity, $m{\cdot}s^{-1}$
U_r	velocity component in radial direction, dimensionless
U_z	velocity component in z-direction, dimensionless
r, z	cylindrical co-ordinates, m

Greek symbols

α	apex angle of cone, *degree*
$\dot{\gamma}$	rate of strain tensor, dimensionless
η	scalar viscosity function, dimensionless
θ	temperature, dimensionless
μ_B	plastic viscosity, $Pa{\cdot}s$
ρ_∞	density of the fluid, $kg{\cdot}m^{-3}$
τ_0	fluid yield stress, Pa

Subscripts

∞	inlet condition
s	cone surface condition

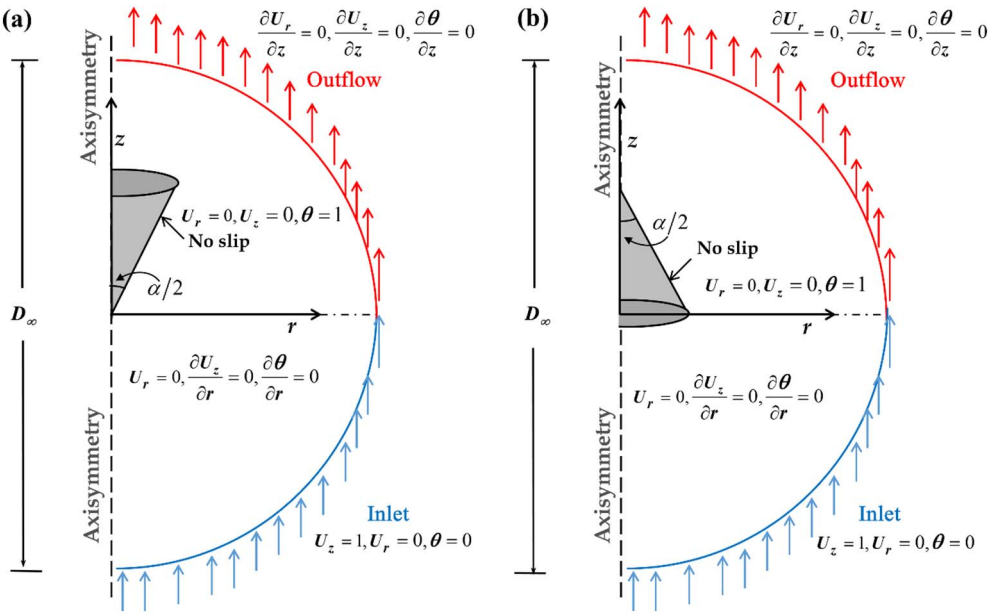


Fig. 1. Schematic of physical domains (a) configuration 1 and (b) configuration 2.

dimensional forms as: $\nabla . \boldsymbol{U} = 0; (\boldsymbol{U}\cdot\nabla)\boldsymbol{U} = -\nabla P + (\nabla\cdot\tau)/Re$; and $(\boldsymbol{U}\cdot\nabla)\theta = (\nabla^2\theta)/(Re\cdot Pr)$; as detailed in our recent studies [3,4,9]. These equations have been rendered dimensionless using D and U_∞ as the linear and velocity scales. Suffice it to mention here that in the present study the modified Bingham plastic fluid viscosity ($\eta_p = 1 + Bn[(1 - \exp(-m\,|\dot{\gamma}|))/|\dot{\gamma}|]$) by the Papanastasiou regularization approach (with, $m = 10^5$) has been employed [13]. Admittedly, some recent studies have obviated the need for a regularization approach [14,15] but the two approaches yield quite similar results.

In this work, the unconfined flow condition is reached by surrounding the cone in a large concentric spherical envelope of fluid of large diameter D_∞, as shown schematically in Fig. 1. The boundary conditions used in the present case are that of no slip on the solid cone (isothermal), uniform flow at the inlet and the standard outflow condition at the outlet [3,4,9].

Dimensional analysis of the field equations suggests the value of the Nusselt number ($Nu_{avg} = hD/k$)to be a function of Re, Bn, Pr and α for a fixed orientation. The present numerical results are studied in terms of isotherms, local (Nu_L) and average Nusselt number (Nu_{avg}) in the ensuing sections.

3. Numerical solution procedure and choice of computational parameters

The numerical solution procedure employed here is identical to that used in our recent study [9]. The numerical mesh consists of very fine triangular cells near the surface of the cone and non-uniform quadrilateral cells in the remaining region (a typical mesh is shown in Fig. 2). The region close to the cone is meshed using free triangular cells where the minimum mesh size (δ/D) is 0.004 and growth rate parameter is 1.1 for the apex downward cone and the corresponding values for apex upward case are 0.006 and 1.1 respectively. The region away from the cone surface was meshed using non-uniform quadrilateral cells with grid expansion ratio as 1.026 for both configurations. A relative

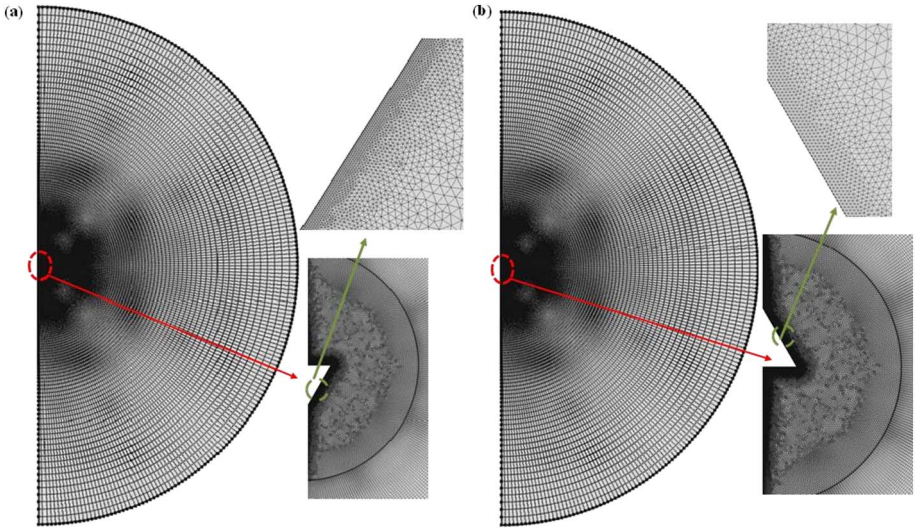


Fig. 2. Typical grid structure for (a) configuration 1 and (b) configuration 2.

convergence criterion of 10^{-5} for all primitive variables was used and the yield surfaces were located using the von Mises yield criterion with a tolerance level of 10^{-5} [16]. A domain size of $D_\infty/D = 400$, a computational mesh (half domain) consisting of 89,384 elements for the configuration shown in Fig. 1(a) and 47623 elements for the orientation of Fig. 1(b) respectively were found to be adequate for the present results to be free from numerical artefacts. Furthermore, a few time-dependent simulations (full domain) were carried out at $Bn = 0$, $Re = 100$ and $Pr = 100$ to confirm the flow and temperature fields to be steady and axisymmetric. For each simulation, the values of Re and Pr are fixed and the values of Bn are input via the parametric sweep feature of COMSOL Multiphysics® (Version 4.3 a). For a 16 GB RAM and 64-bit Windows 7 operating system with an Intel® Core™ i7-3370 CPU @ 3.40 GHz processor, the typical simulation time varies from 1.5 to 3 h depending upon the values of Re, Pr and α.

4. Results and discussion

The heat transfer results are analyzed here in terms of isotherms, local and average Nusselt number over the range of conditions: $1 \le Re \le 100$, $10 \le Pr \le 100$, $0 \le Bn \le 100$ and cone angle, $20^\circ \le \alpha \le 150^\circ$. Initially, the effect of cone orientation and of Re, Pr and Bn is examined for a fixed value of cone angle i.e., $\alpha = 60^\circ$ and then further results are presented to explore the effect of the cone angle. At the outset, it is worthwhile to demonstrate the reliability and the accuracy of the new numerical results obtained in this work. As extensive validations with the experimental results on drag in Newtonian fluids and Bingham plastic fluids for both orientations are reported in ref. [9], additional comparisons were made for free convection in Bingham plastic fluid in a 2-D duct and with the laminar lid-driven cavity flow. In both cases, the present results were within ± 0.5% of the literature values [17,18]. Based on the results reported in ref. [9] coupled with the additional comparisons, the new numerical results presented here are considered to be reliable to within ~2%. Also, since the fluid mechanical aspects including the size and shape of the yielded and unyielded regions are discussed in detail in ref. [9], consideration is given mainly to the corresponding heat transfer results here. Broadly, there were two sub-domains of unyielded material: small regions at the stagnation points (as mentioned below) and the faraway material in a piston-like motion. As expected, the fluid-like cavity encapsulating the cone is like an apple with slightly flattened top and bottom. With the increasing Bn, this region shrank while it expanded with Re, but its radius is roughly of the order of 2–3 times the cone radius.

4.1. Isotherm contours

Representative isotherm contours are shown in Fig. 3 for $\alpha = 60^\circ$ for a range of values of Re and Pr. Since qualitatively similar patterns are

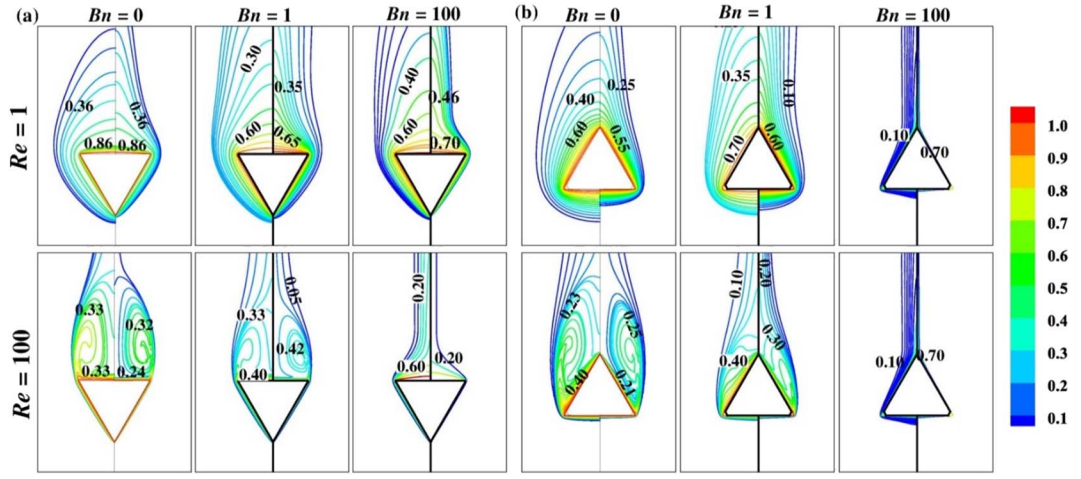


Fig. 3. Isotherm contours in the vicinity of the cone ($\alpha = 60^\circ$) for (a) configuration 1 (b) configuration 2 for $Pr = 10$ (left half) and $Pr = 100$ (right half).

obtained for the other values of α, these are not presented here. In Newtonian fluids ($Bn = 0$), the dominant role of conduction at low $Re = 1$ is evident for both orientations. At $Re = 100$, there exists a wake for both orientations and this is reflected here in the form of the recirculation region which promotes heat transfer with the increasing Reynolds number. However, the vortex forms in the rear for the apex downward, due to the diverging flow area in the rear because of the establishment of an adverse pressure gradient. The wake grows in both directions with the increasing Reynolds number. In the other orientation, the flow separates along the slanted surface, due to the sudden bending of the flow at the bottom of the cone and in this case also, the wake length increases with the increasing Reynolds number. In a sense, the latter flow is more like that normal to a circular disk. These recirculating regions augment heat transfer. However, as soon as the fluid yield stress is introduced, the recirculation region is suppressed, for instance note the results for $Bn = 1$. This leads to suppression in the extent of fluid-like regions which has an adverse influence on heat transfer. For the apex downward cone, there is a blob of unsheared material adhering at the rear stagnation point which grows in size in both directions with the increasing Bn. On the other hand, in the case of an upward cone, there is a very small amount of unsheared fluid at the front stagnation point which grows a little with the increasing Bn and it diminishes with the increasing Re. Thus, the temperature field in the vicinity is determined by the size of the recirculation region and of the unsheared material adhering at the stagnation points in the two configurations. As expected, at very high values of yield stress (e.g., see the results for $Bn = 100$), fluid-like layer is extremely thin adjacent to the cone and the yield surfaces do not undergo any further changes (corresponding to the fully plastic limit). Thus, all in all, one can postulate a positive effect of Re, Pr and Bn on heat transfer for a given geometry (value of α) and the orientation of the cone in general.

4.2. Distribution of Local Nusselt number on the surface of cone

Typical local Nusselt number distributions for the two orientations are shown in Fig. 4 for extreme values of the Reynolds and Prandtl numbers and cone angle, $\alpha = 60^{\circ}$. Unlike in the case of a Newtonian fluid, the viscosity of a Bingham fluid varies along the surface and hence the local values of the Re, Pr and Bn also deviate from their nominal values. For a downward cone, the maximum Nusselt number is obtained at the tip and it gradually decreases along the inclined surface, exhibiting another peak at point B because of the sharp bending of the isotherms to follow the body contour and again decreasing along the flat base, akin to the boundary layer flow over a flat surface and also due to the unsheared material present at the rear stagnation point. At low Re, advection is weak and the flow remains attached and therefore no recovery is seen due to the absence of the wake. Since the fluid yield stress further suppresses flow separation, the recovery in the Nusselt number also diminishes. However, the formation of the unyielded regions further interferes with this effect and eventually the impact of the fluid yield stress is seen to approach a saturation limit independent of the values of Re and Pr.

For the opposite orientation, where the base of the cone is facing the oncoming stream, the Nusselt number is seen to be minimum at the front stagnation point due to the unyielded fluid present here, whereas the maximum value occurs at the edge due to the severe bending of isotherms and it exhibits another peak at the tip of the cone. At low Pe ($= Re \cdot Pr$), the fluid advection is weak and there is only a small increase in heat transfer along the flat base and the inclined surface of the cone with Bn and Pr; however, at $Pe = 100$ ($Re = 1$, $Pr = 100$), the influence

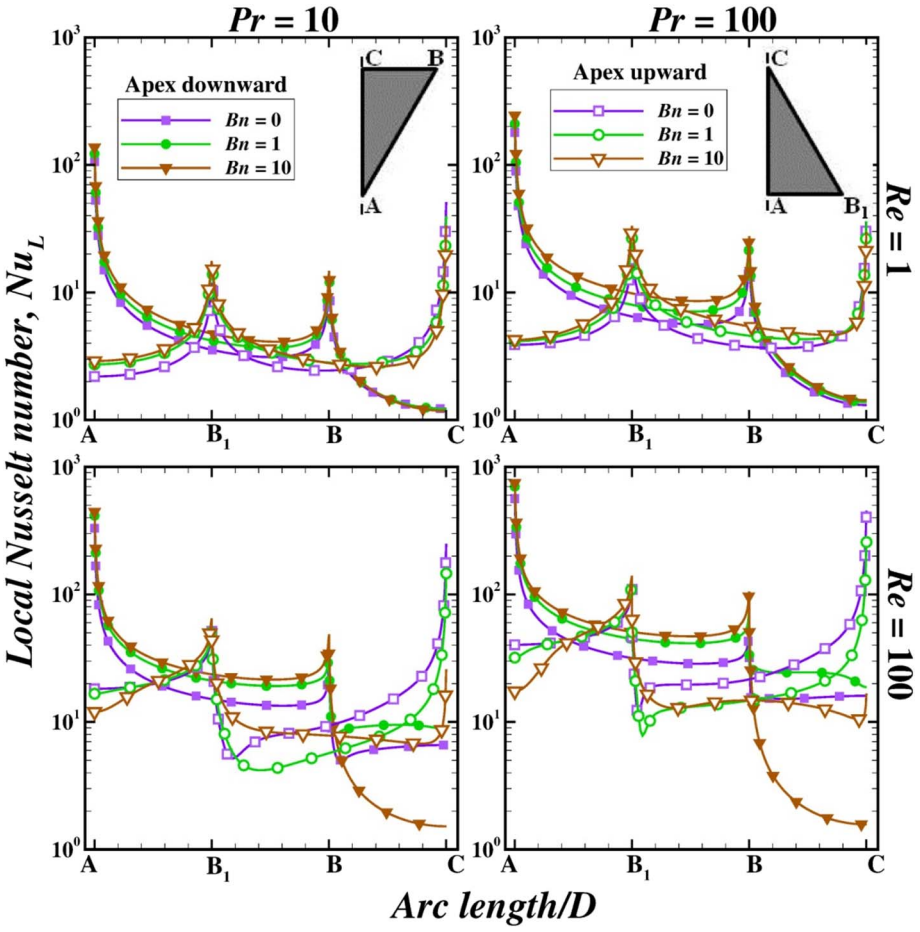


Fig. 4. Variation of the local Nusselt number along the surface of cone ($\alpha = 60^{\circ}$) for (a) configuration 1 and (b) configuration 2.

of *Bn* along the inclined surface is complicated. This is a consequence of the two competing effects arising from the thinning of the fluid-like (yielded) region which tends to enhance the heat transfer. However, at the same time, there is a concomitant expansion of the unyielded regions which increase the conduction resistance (depending upon the portion of the cone surface covered with the unyielded material). This interaction is further accentuated at $Re = 100$. From another vantage point, the trends seen at $Re = 100$ in Fig. 4 stem from the combined effects of the three boundary layers (momentum, thermal and stress) along the slanted surface. The thickness of these three boundary layers scale differently with *Re*, *Bn* and *Pr*. Thus, for instance, it is conceivable that for a specific combination of the values of the influencing parameters *Re*, *Pr* and *Bn*, the fluid may be yielded everywhere within the thermal boundary layer while it might not be so for other combinations of conditions.

4.3. Average Nusselt number

The average Nusselt number, Nu_{avg} is expected to be a function of *Re*, *Pr*, *Bn*, α and its orientation. This multivariable functional dependence is shown in Fig. 5 for $\alpha = 60^{\circ}$ and two orientations. At low *Pe* ($Re = 1$, $10 \le Pr \le 100$), Bingham number has a positive impact on the average Nusselt number (see Fig. 5) and this is intensified at high Prandtl numbers (at $Re = 1$). At $Pe = 10$, heat transfer occurs mainly by conduction and hence the orientation has no effect. As the fluid advection increases, the effect of orientation begins to manifest itself, i.e., Nu_{avg} for configuration 1 is slightly higher than that for configuration 2. This is clearly due to the differences in the flow pattern in terms of the unsheared material vis-à-vis the size of the recirculating regions [9]. At $Re = 20$, not only the difference between the two orientations is magnified but also the two values differ increasingly as the Prandtl number is incremented. At high $Re = 50$ and $Re = 100$, the results are seen to be quite different, in line with the local Nusselt number results seen in Fig. 4. Naturally, the inertial forces at $Re = 50$ and $Re = 100$ tend to promote fluid yielding which is resisted by *Bn*. The minimum value of the Nusselt number denotes the equilibrium between these two effects. At $Re = 50$, this occurs at $Bn = 3$ for configuration 1 whereas there is no such minimum observed for configuration 2 because the unyielded zone is much smaller in this case than that in the former. Similarly at $Re = 100$, for configuration 1, the minimum value occurs at about $Bn = 6$ whereas it occurs at $Bn = 2$ for configuration 2. Finally, Table 1 shows the effect of cone angle on the average Nusselt number for a few cases. For both configurations, the average Nusselt number displays an increasing trend upto a certain value of α, beyond which it shows a downward trend for fixed values of *Re*, *Bn* and *Pr*. The increase in Nusselt number of course is dependent on the values of *Bn*, *Re*, *Pr* and orientation. It ranges from ~ 2% to 140%, though it is only quite significant in the range $20^{\circ} \le \alpha \le 90^{\circ}$. Also, the present results are pretty insensitive to the value of $Bn > 100$ due to the fully plastic flow limit. Finally, it is desirable to develop simple expressions for average Nusselt number. Following the approach used in [3,4,9], the modified forms of the Reynolds number ($Re^{*} = Re/(1 + Bn)$) and Prandtl number ($Pr^{*} = Pr(1 + Bn)$) are introduced to define the Colburn factor, $j_H = Nu_{avg}/(Re^{*} \cdot Pr^{*1/3})$. This functional relationship is well described as:

$$j_H = a(2 + \cos(\alpha))^b/(Re^{*})^c \quad (1)$$

The numerical values of *a*, *b* and *c* and the mean and maximum deviations (for 1830 individual data points for each orientation) are summarized in Table 2. Only about 10 data points for each orientation

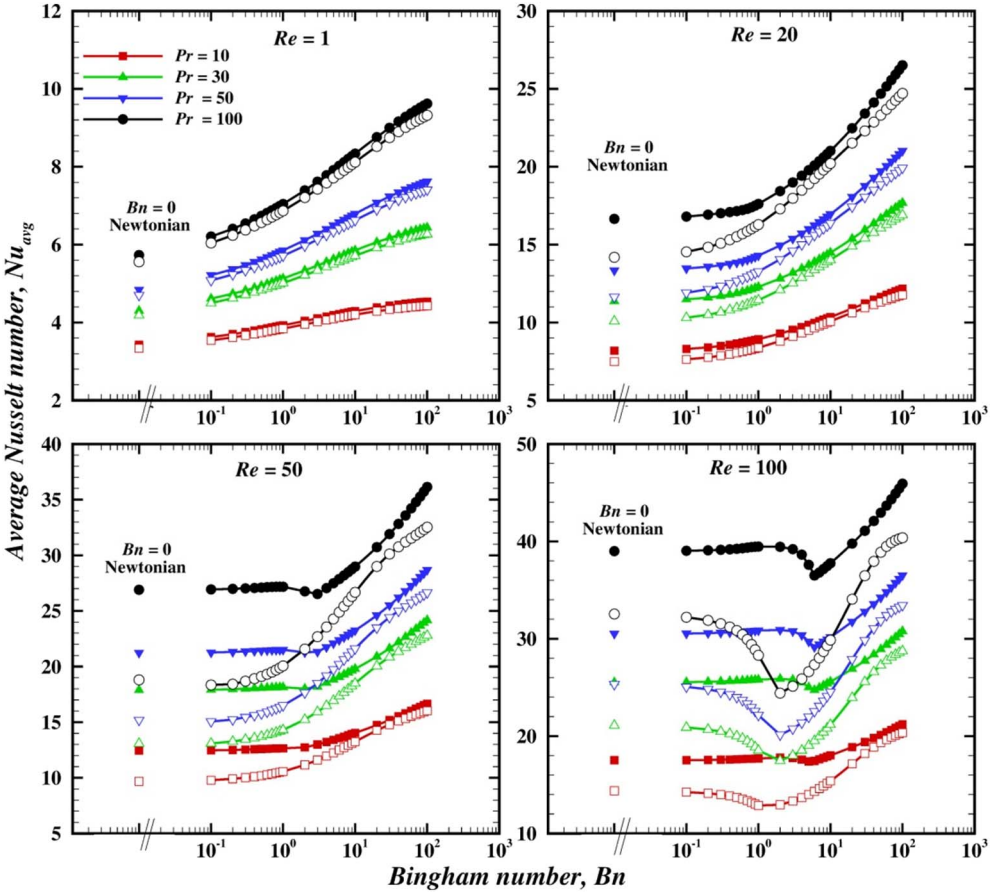


Fig. 5. Dependence of the average Nusselt number on the Reynolds number, Prandtl number and Bingham number for configuration 1(filled symbols) and configuration 2(unfilled symbols) for ($\alpha = 60^{\circ}$).

Table 1
Effect of cone angle on the average Nusselt number for (a) configuration 1 (b) configuration 2.

(a) Configuration 1 (*apex downward*)

Re	*Bn*	$\alpha = 20^\circ$	$\alpha = 40^\circ$	$\alpha = 50^\circ$	$\alpha = 60^\circ$	$\alpha = 90^\circ$	$\alpha = 120^\circ$	$\alpha = 150^\circ$
$Pr = 10$								
1	0	2.6626	3.1565	3.3080	3.4262	3.5925	3.6224	3.5716
	1	3.0787	3.6319	3.7989	3.9274	4.0981	4.1111	4.0345
	10	3.3020	3.9453	4.1417	4.2926	4.5026	4.5197	4.4238
	50	3.3917	4.0935	4.3117	4.4808	4.7289	4.7623	4.6595
	100	3.4149	4.1343	4.3591	4.5336	4.7941	4.8335	4.7290
100	0	11.665	15.292	16.545	17.521	19.355	20.048	19.966
	1	12.063	15.555	16.754	17.703	19.504	20.194	20.096
	10	13.713	16.599	17.436	18.018	18.635	18.280	17.700
	50	15.964	18.831	19.594	20.122	20.596	19.987	18.613
	100	17.001	20.018	20.746	21.220	21.468	20.619	18.874
$Pr = 100$								
1	0	4.4112	5.2956	5.5478	5.7303	5.9230	5.8523	5.6546
	1	5.5378	6.5617	6.8441	7.0429	7.2129	7.0597	6.7526
	10	6.5415	7.7669	8.1016	8.3355	8.5396	8.3219	7.8744
	50	7.1647	8.6086	9.0046	9.2811	9.5460	9.3059	8.7446
	100	7.3639	8.9050	9.3268	9.6203	9.9091	9.6589	9.0519
100	0	24.504	33.351	36.614	38.998	43.451	45.488	45.562
	1	25.316	34.006	37.014	39.465	44.357	46.349	46.375
	10	28.582	34.815	36.508	37.754	38.779	38.683	39.606
	50	34.198	40.405	41.822	42.940	43.267	41.426	37.860
	100	37.126	43.672	44.939	45.927	45.419	42.747	37.920

(b) Configuration 2 (*apex upward*)

Re	*Bn*	$\alpha = 20^\circ$	$\alpha = 40^\circ$	$\alpha = 50^\circ$	$\alpha = 60^\circ$	$\alpha = 90^\circ$	$\alpha = 120^\circ$	$\alpha = 150^\circ$
$Pr = 10$								
1	0	2.6734	3.1376	3.2817	3.3415	3.5535	3.5981	3.5590
	1	3.1016	3.6237	3.7818	3.8423	4.0599	4.0851	4.0201
	10	3.3279	3.9361	4.1224	4.1994	4.4588	4.4882	4.4059
	50	3.4190	4.0799	4.2868	4.3788	4.6781	4.7252	4.6398
	100	3.4410	4.1190	4.3319	4.4277	4.7404	4.7944	4.7082
100	0	9.6962	11.148	13.089	14.378	17.143	18.603	19.239
	1	10.574	11.728	12.134	12.867	16.709	18.522	19.297
	10	13.432	15.277	15.538	15.391	15.527	15.915	16.080
	50	15.687	18.720	19.347	19.324	19.157	17.713	17.019
	100	16.608	19.741	20.432	20.350	20.536	18.837	17.184
$Pr = 100$								
1	0	4.4304	5.2453	5.4777	5.5593	5.8146	5.7708	5.6083
	1	5.5746	6.5326	6.7942	6.8667	7.1099	6.9754	6.6994
	10	6.5741	7.7235	8.0352	8.1173	8.4144	8.2184	7.8094
	50	7.2577	8.5367	8.9070	9.0096	9.3884	9.1795	8.6745
	100	7.4108	8.8198	9.2136	9.3201	9.7350	9.5217	8.9770
100	0	18.250	23.976	29.198	32.540	38.962	42.493	43.931
	1	20.465	21.781	23.140	28.301	38.644	43.067	44.771
	10	27.332	30.724	30.790	29.861	29.803	31.261	32.644
	50	32.762	38.962	40.049	38.885	38.112	33.850	32.681
	100	34.492	40.953	42.164	40.369	41.016	36.109	31.617

show deviations > 25%, without any discernable trends. Such large deviations denote the inherent difficulty in consolidating the results for Newtonian and Bingham plastic fluids, as is discussed in detail with regard to the experimental and numerical drag results for a sphere [1–3]. However, in the limit of $Bn \to 0$, Re^* and Pr^* reduce to the expected definitions for Newtonian fluids. Also, Eq. (1) retains the widely accepted scaling of $Nu \sim Pr^{1/3}$ and the value of c also does not differ too much from the oft-quoted value of (2/3) in the literature in the context

Table 2
Correlation constants used in Eq. (1).

Range	Configuration 1 (*apex downward*)					Configuration 2 (*apex upward*)				
	a	*b*	*c*	% Error		*a*	*b*	*c*	% Error	
				Mean	Maximum				Mean	Maximum
$Re^* < 1$	2.04	−0.22	0.70	9.29	29.7	1.93	−0.14	0.71	8.31	29.8
$1 \le Re^* \le 100$	1.56	−0.06	0.63	11.4	29.8	1.53	−0.04	0.66	10.8	29.9

of Newtonian fluids. Thus, a Bingham fluid effectively behaves like a Newtonian fluid with very high viscosity.

5. Conclusions

This study elucidates the relationship between heat transfer from an isothermal cone in Bingham plastic fluids in two different orientations over the range as: $1 \le Re \le 100$, $10 \le Pr \le 100$, $0 \le Bn \le 100$ and $20^{\circ} \le \alpha \le 150^{\circ}$. The temperature field and thus the overall heat transfer are determined by the two competing effects, wake characteristics and unyielded material attached at the stagnation points and on the surface of the cone. Thus the average Nusselt number for apex downward configuration is generally higher than those of apex upward configuration. In both cases, the Nusselt number exhibits a positive dependence on Re^{*} and Pr^{*}. Also, the cone angle influences heat transfer in a positive manner. The simple predictive expression presented here can be used to interpolate the present results and/or to estimate the value of the Nusselt number in a new application.

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